

Notes and references of the 2026 CIMPA School

Lucas Cardoso¹ and Gabriel Carvalho¹

Escola Politécnica, Universidade de São Paulo, Brazil
{lcardoso, gcarvalho}@larc.usp.br

Abstract. This notes refer to the two weeks course of the 2026 CIMPA School, held in Santiago, Chile. It contains aspects of lattice-, code-, isogeny-, multivariate-, and MPCitH-based cryptography.

1 Introduction

2 Code-based cryptography

3 Isogeny-based cryptography

4 Lattice-based cryptography

The sessions on lattice-based cryptography was conducted by professor Fernando Virdia and by professor Nicolas Thériault, respectively. The first session was focused on a general overview of the hard problems from Lattice Theory (CVP, SVP, Search LWE, Decision LWE, and others), worst-case to average-case reduction, and cryptanalytic results. The second session focused on the formalization of MLWE, RLWE, ML-DSA, and fully homomorphic encryption.

Problem sets 1 and 2 correspond to the first part of the course, while problem sets 3 and 4 corresponds to the latter part.

4.1 Problem Set 3

Exercise 1 (RLWE vs. LWE). Let $\mathcal{R}_q = \mathbb{Z}_q[x]/f(x)$ with $q = 101$ and $f(x) = x^5 - 1$.

- Construct the system $\mathbf{b} = \mathbf{A} \cdot \mathbf{s} + \mathbf{e}$ associated to

$$a(x) = 83x^4 + 32x^3 + 71x^2 + 54x + 9$$

for the polynomials $s(x) = s_0 + s_1x + s_2x^2 + s_3x^3 + s_4x^4$ and $e(x) = e_0 + e_1x + e_2x^2 + e_3x^3 + e_4x^4$.

- Consider the same problem, but with $f(x) = x^5 + 3x^4 + x^2 + 7$.
- Consider the same problem (back to $f(x) = x^5 - 1$), but with

$$a(x) = 31x^4 + 49x^3 + 16x^2 + 8x + 1$$

Exercise 2 (Growth factor). For a polynomial $f(x)$ of degree d , we can define the growth factor τ as

$$\tau = \sup \left\{ \frac{\max |c_i|}{\max |a_i| \cdot \max |b_i|} \right\}$$

where

$$a(x) = \sum_{0 \leq i < d} a_i x^i, b(x) = \sum_{0 \leq i < d} b_i x^i, \text{ and } c(x) = \sum_{0 \leq i < d} c_i x^i \equiv a(x) \cdot b(x) \pmod{f(x)}.$$

In the following, assume that α is relatively small.

- Show that both $f(x) = x^d - 1$ and $f(x) = x^d + 1$ have growth factor $\tau = 1$.
- Show that both $f(x) = x^d - \alpha$ has growth factor $\tau = |\alpha|$.
- Show that both $f(x) = x^d - x - \alpha$ and $f(x) = x^d + x - \alpha$ have growth factor $\tau = |\alpha| + 1$.
- What is the growth factor $f(x) = x^d - \alpha x + 1$?

Solution 1.

Exercise 3 (Bounding the noise). Let $\mathcal{R}_q = \mathbb{Z}_q[x]/f(x)$ with $f(x)$ a polynomial of degree d and growth factor τ . Let $\mathcal{M}$ be a n -dimensional module over $\mathcal{R}_q$. Suppose that all polynomial entries in $\mathcal{R}_q$ have coefficients bounded by b . Show that all coefficients of c are bounded by

$$b + 2\tau n d b^2$$

Exercise 4 (Baby ML-KEM). We define a small ML-KEM setting. Let $q = 1009$, $f(x) = x^4 - 1$, $d = 1$, $K = 500$,

$$\mathbf{PK}_1 = \mathbf{A} = 798x^3 + 141x^2 + 597x + 894$$

and assume all “small” coefficients are in $\{0, 1, 2, 3, 4\}$.

- Show that $\mathbf{pk}_0 = 817x^3 + 631x^2 + 765x + 81$ corresponds to a possible public key for $\mathbf{r} = 3x^3 + x^2 + 4x$.
- Obtain the message encrypted as

$$\begin{cases} \mathbf{ct}_0 = 174x^3 + 365x^2 + 212x + 212 \\ \mathbf{ct}_1 = 718x^3 + 810x^2 + 22x + 150 \end{cases}$$

- Can you tell which temporary secret $\mathbf{s}$ was used?

4.2 Problem Set 4

5 MPCitH-based cryptography

6 Multivariate-based cryptography

6.1 Problem Set 2

Exercise 5. Prove that the polar form is bilinear and symmetric in the following cases:

- $\mathcal{F}$ is quadratic and homogeneous.
- $\mathcal{F}$ is quadratic and inhomogeneous.
- In the homogeneous case, if M is the symmetric matrix such that $\mathcal{F}(\mathbf{x}) = \mathbf{x}^\top \mathbf{M} \mathbf{x}$, show that the polar form is $\mathcal{G}(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{M} \mathbf{y}$.

Solution 2. Write $\mathcal{F}(\mathbf{x}) = \mathbf{x}^\top \mathbf{M} \mathbf{x}$. From the definition of polar form, we have $\mathcal{G}(\mathbf{x}, \mathbf{y}) = \mathcal{F}(\mathbf{x} + \mathbf{y}) - \mathcal{F}(\mathbf{x}) - \mathcal{F}(\mathbf{y})$. Substituting $\mathbf{x}$ by $\alpha \mathbf{x}$ for $\alpha \in \mathbb{F}_q$, we have:

$$\mathcal{G}(\alpha \mathbf{x}, \mathbf{y}) = \mathcal{F}(\alpha \mathbf{x} + \mathbf{y}) - \mathcal{F}(\alpha \mathbf{x}) - \mathcal{F}(\mathbf{y})$$

For item b), write $\mathcal{F}(\mathbf{x}) = \mathbf{x}^\top \mathbf{M} \mathbf{x} + f^\top \mathbf{x} + c$.

Solution 3. We have that

$$\binom{2q}{q+2} = \frac{(2q)!}{(q+2)!(q-2)!} \approx \sqrt{\frac{2q}{\pi(q^2-4)}} \frac{(2q)^{2q}}{e^{2q}} \frac{e^{q+2} e^{q-2}}{(q+2)^{q+2} (q-2)^{q-2}}$$

The probability of finding $q+2$ valid transcripts out of $2q$ transcripts in total (because the challenge space is $\mathcal{CH} = \mathbb{F}_q \times \{0, 1\}$) is given by

$$\Pr[x = q+2] = \binom{2q}{q+2} p^{q+2} (1-p)^{2q-(q+2)}$$

where $p = \frac{1}{2} + \frac{1}{2q} + \varepsilon$

7 Conclusion